Business Question and Visualization Report

Date	28 October 2026
Team ID	SKILLWALLET-EV-01
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

Implemented Business Questions and Visualizations in Tableau:

1. What are the global annual sales trends across different EV manufacturers?

• Visualization: Horizontal Bar Chart ranking top EV manufacturers by Annual Unit Sales.
• Key Insight: Insight: Identifies leading automotive brands dominating consumer adoption across regional markets.

[Visualization Implemented in Tableau Workbook & Published Dashboard]

2. How does real-world driving range correlate with battery capacity across vehicle categories?

• Visualization: Scatter Plot showing Battery Capacity (kWh) vs. Driving Range (km) colored by Vehicle Category.
• Key Insight: Insight: Demonstrates a strong positive linear relationship, with sports cars and luxury sedans delivering >550 km range.

[Visualization Implemented in Tableau Workbook & Published Dashboard]

3. How do DC fast-charging times compare across different battery chemistries?

• Visualization: Box-and-Whisker / Column Chart comparing Fast Charging Time (min) by Battery Type (LFP, NMC, NCA, Solid-State).
• Key Insight: Insight: Advanced battery chemistries reduce charging duration under 25 minutes, addressing consumer range anxiety.

[Visualization Implemented in Tableau Workbook & Published Dashboard]

4. What is the relationship between motor power (kW) and energy efficiency (Wh/km)?

• Visualization: Dual-Axis Line and Bar Chart evaluating Motor Power vs. Efficiency across Drivetrain configurations.
• Key Insight: Insight: Dual-motor AWD setups consume higher energy per km but provide superior acceleration (0-100 km/h in <4.5 sec).

[Visualization Implemented in Tableau Workbook & Published Dashboard]

5. How does vehicle price impact the 3-year resale value percentage?

• Visualization: Scatter Plot with Trend Line mapping Vehicle Price (USD) against 3-Year Resale Value (%).
• Key Insight: Insight: Mainstream utility EVs in the $30k–$50k segment retain highest resale value retention (~58%).

[Visualization Implemented in Tableau Workbook & Published Dashboard]

6. How are yearly insurance costs distributed across vehicle categories?

• Visualization: Grouped Bar Chart detailing Annual Insurance Cost (USD) across Categories and Safety Ratings.
• Key Insight: Insight: Higher safety rating stars (5-star) significantly reduce annual insurance premiums despite high motor power.

[Visualization Implemented in Tableau Workbook & Published Dashboard]

7. What is the total annual CO2 emissions saved per market region?

• Visualization: Geographic Map / Treemap displaying CO2 Saved (tons/year) by Market Region.
• Key Insight: Insight: High-volume adoption regions like Europe and North America prevent millions of metric tons of greenhouse emissions yearly.

[Visualization Implemented in Tableau Workbook & Published Dashboard]

8. How does battery degradation over 5 years vary by manufacturing country and cycle life?

• Visualization: Heatmap / Matrix comparing 5-Year Degradation (%) against Cycle Life across manufacturing origins.
• Key Insight: Insight: Identifies optimal manufacturing quality and battery longevity standards.

[Visualization Implemented in Tableau Workbook & Published Dashboard]

9. What is the overall customer rating breakdown across recommended EV use-cases?

• Visualization: Radar / Stacked Bar Chart evaluating Customer Ratings for Urban Commute, Long-Distance, and Commercial fleets.
• Key Insight: Insight: Urban commute models score highest in customer satisfaction due to lower operational costs and compact form factors.

[Visualization Implemented in Tableau Workbook & Published Dashboard]

Live Tableau Public Dashboard Reference:
https://public.tableau.com/views/EV_Market_Performance_17931262572980/evmarketperformancedashboard